



ADBMS_PROJECT_REPORT

E-COMMERCE SYSTEM



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PROJECT REPORT

E-COMMERCE MANAGEMENT SYSTEM USING JAVA SWING AND MONGODB

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Course: Advanced Database Management Systems (ADBMS)

CS-IV, SECTION-D

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1. Introduction

E-commerce systems are widely used to manage products, customers, orders, and transactions in an efficient manner. Manual management of such data can be difficult and may lead to errors. Therefore, a computerized system is required to automate these activities and improve efficiency.

This project presents an E-Commerce Management System developed using Java Swing and MongoDB. The system provides a graphical user interface through which users can manage products, customers, shopping carts, and orders. MongoDB is used as the backend NoSQL database to store and retrieve data efficiently.

The project demonstrates how a desktop application can be integrated with a NoSQL database to perform different database operations in a real-world environment.

2. Objectives

The main objectives of this project are:

- To develop an E-Commerce Management System using Java Swing.
- To use MongoDB as the backend NoSQL database.

- To implement CRUD operations.
 - To manage products, customers, orders, and cart records.
 - To understand the practical use of MongoDB in application development.
 - To provide a simple and user-friendly interface.
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3. Scope of the System

The system provides the following functionalities:

- Product Management
- Customer Management
- Shopping Cart Management
- Order Management
- Product Search
- Inventory Tracking
- MongoDB Database Integration

The system can be used to manage product information, customer records, shopping carts, and order processing in a centralized environment.

4. Technology Stack

Technology	Purpose
Java	Programming Language
Java Swing	Graphical User Interface
MongoDB	NoSQL Database
MongoDB Java Driver	Database Connectivity
Maven	Dependency Management
IntelliJ IDEA	Development Environment

5. Database Design

The database used in this project is named **EcommerceDB**.

The system consists of four collections:

- customers
 - products
 - orders
 - cart
-

5.1 Customers Collection

The customers collection stores information about registered customers.

Sample Document

```
{
  "_id": "C001",
  "name": "Shifa",
  "email": "shifa@gmail.com",
  "password": "1234"
}
```

This collection currently contains customer records such as Shifa, Alia, Sanaullah, Ahmed, and Fatima.

5.2 Products Collection

The products collection stores information about products available in the system.

Sample Document

```
{
  "_id": "P001",
  "name": "Laptop",
  "category": "Electronics",
  "price": 85000,
  "stock": 10
}
```

The collection contains products from multiple categories including:

- Electronics
- Accessories
- Furniture

Examples include laptops, smartphones, headphones, keyboards, gaming chairs, study tables, and power banks.

5.3 Orders Collection

The orders collection stores information about customer purchases.

Sample Document

```
{
  "_id": "O001",
  "customerId": "C001",
  "productId": "P001",
  "quantity": 1,
  "total": 85000,
  "status": "Delivered"
}
```

The system maintains different order statuses such as Pending, Shipped, and Delivered.

5.4 Cart Collection

The cart collection stores products selected by customers before placing orders.

Sample Document

```
{
  "_id": "CRT001",
  "customerId": "C001",
  "productId": "P005",
  "quantity": 2
}
```

This collection helps customers manage selected products before final checkout.

6. Dataset Description

The database contains sample records for testing and demonstration purposes.

Dataset Summary

Collection	Records
Customers	5
Products	12
Orders	5
Cart	3

Product Categories

- Electronics
- Accessories
- Furniture

The dataset was designed to simulate a real e-commerce environment and test system functionalities effectively.

7. Functionalities Implemented

Admin Module

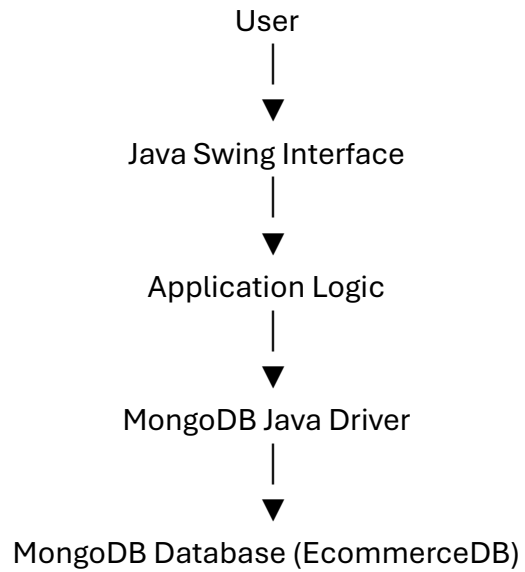
- Add Product
- Search Product
- View Products
- Monitor Orders
-

Customer Module

- Add Customer
- View Products

- Add Items to Cart
- View Cart
- Place Orders

8. System Architecture



9. CRUD Operations

Create

- Add Customer
- Add Product
- Create Order
- Add Item to Cart

Read

- View Products
- Search Products
- View Orders
- View Cart

Update

- Update Product Information
- Update Product Stock
- Update Order Status

Delete

- Remove Product Records
- Remove Cart Records
- Delete Existing Data

10. Implementation Analysis

The system successfully integrates Java Swing with MongoDB to perform database operations through a graphical user interface.

MongoDB provides efficient storage and retrieval of information, while Java Swing enables users to interact with the system using forms, tables, buttons, and dialogs.

The use of separate collections for customers, products, orders, and cart records improves data organization and simplifies management.

11. Advantages of MongoDB

- Flexible document-based structure
 - Easy integration with Java
 - Fast data retrieval
 - High scalability
 - Reduced database complexity
 - Suitable for modern applications
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12. Future Enhancements

The system can be further improved by adding:

- User Authentication
 - Online Payment Gateway
 - Sales Reports
 - Product Categories
 - Inventory Analytics
 - Mobile Application Support
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13. Conclusion

The E-Commerce Management System successfully demonstrates the implementation of a NoSQL database using MongoDB. The system provides functionalities for managing customers, products, shopping carts, and orders through a user-friendly graphical interface developed using Java Swing.

The project fulfills the requirements of the Advanced Database Management Systems course by implementing CRUD operations and integrating a NoSQL database with a desktop application. The system is efficient, easy to use, and can be expanded with additional features in the future.

Final Statement

All CRUD operations (Create, Read, Update, Delete) were implemented using MongoDB, fulfilling the requirements of the ADBMS project.

https://github.com/AliaKalwar/E-COMMERCE_SYSTEM